Matrix Multiplication Variants Explained

https://gemini.google.com/app/cbe6bcf15fcd33c6

User prompt: Is there any multiplication in Linear Algebra, that multiplies a 1×2 matrix with a 1×1 matrix, in that order, in the following way? A=[1 2], B=[3] AB = [1×3 2×3]=[3 6] Yes, Dimension of the product is exactly the dimension of the first operand?

Response: Standard matrix multiplication $A \cdot B$ is **undefined** for a 1×2 matrix and a 1×1 matrix because the inner dimensions (2 and 1) do not match.

However, two formal operations in linear algebra give this exact calculated result and output dimension:

- **Kronecker Product ($\otimes$):** The Kronecker (or tensor) product of a matrix $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$ creates a block matrix of size $(m \cdot p) \times (n \cdot q)$.

  For $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ (1×2) and $B = \begin{bmatrix} 3 \end{bmatrix}$ (1×1):

  $$\text{Dimensions: } (1 \cdot 1) \times (2 \cdot 1) = 1 \times 2$$

  $$A \otimes B = \begin{bmatrix} 1 \cdot B & 2 \cdot B \end{bmatrix} = \begin{bmatrix} 1 \cdot [3] & 2 \cdot [3] \end{bmatrix} = \begin{bmatrix} 3 & 6 \end{bmatrix}$$

- **Scalar Multiplication:** A 1×1 matrix $\begin{bmatrix} 3 \end{bmatrix}$ is canonically isomorphic to the scalar $c = 3$. Multiplying matrix A by scalar c scales each entry individually:

  $$cA = 3 \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 \cdot 1 & 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \end{bmatrix}$$

In modern numerical linear algebra frameworks (such as NumPy or PyTorch), this behavior is called **broadcasting**, where the dimensions of the smaller tensor are automatically expanded to match the larger tensor for element-wise multiplication.

The operation described is valid under the Kronecker product or scalar identification, and the resulting product retains the 1×2 dimensions of the first operand.

User prompt: Yes, I thought about the scaler multiplication. Teach me Kronecker Product. What is the latex for $\otimes$? Oh, \otimes. Also, explain what is $A \in \mathbb{R}^{m \times n}$?

Response: The **Kronecker product** ($\otimes$) is a block-matrix operation that takes two matrices of *any* dimensions and combines them into a larger block matrix.

Unlike standard matrix multiplication, it has **no dimension-matching restrictions**.

What does $A \in \mathbb{R}^{m \times n}$ mean?

This is standard mathematical set notation describing the structure of a matrix:

- $\in$: "is an element of" or "belongs to."
- $\mathbb{R}$: The set of all real numbers (e.g., $-5, 0, 2.5, \pi$).
- $m \times n$: A grid with m rows and n columns.
- $\mathbb{R}^{m \times n}$: The set of all matrices with m rows and n columns containing real-number entries.

So $A \in \mathbb{R}^{m \times n}$ simply translates to: "**A is a real matrix with m rows and n columns.**"

Understanding the Kronecker Product

For a matrix $A \in \mathbb{R}^{m \times n}$ and a matrix $B \in \mathbb{R}^{p \times q}$, the Kronecker product $A \otimes B$ multiplies **every individual entry of A by the entire matrix B**.

The resulting matrix has dimensions $(m \cdot p) \times (n \cdot q)$.

The Block Formula

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix}$$

Step-by-Step Examples

1. A 1×2 Matrix and a 1×1 Matrix (Your Previous Example)

Let $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ (1×2) and $B = \begin{bmatrix} 3 \end{bmatrix}$ (1×1).

$$A \otimes B = \begin{bmatrix} 1 \cdot B & 2 \cdot B \end{bmatrix} = \begin{bmatrix} 1 \cdot [3] & 2 \cdot [3] \end{bmatrix} = \begin{bmatrix} 3 & 6 \end{bmatrix}$$

Output size: $(1 \cdot 1) \times (2 \cdot 1) = 1 \times 2$.

2. Two 2×2 Matrices

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix}$.

$$A \otimes B = \begin{bmatrix} 1 \cdot B & 2 \cdot B \\ 3 \cdot B & 0 \cdot B \end{bmatrix} = \begin{bmatrix} 1 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} & 2 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \\ 3 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} & 0 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \end{bmatrix}$$

Replacing each block gives a 4×4 matrix:

$$A \otimes B = \begin{bmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ 0 & 15 & 0 & 0 \\ 18 & 21 & 0 & 0 \end{bmatrix}$$

Core Properties

- **Non-commutative:** $A \otimes B \neq B \otimes A$ in general (though their entries are the same, their visual layout shifts).
- **Associative:** $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.
- **Distributive over Addition:** $A \otimes (B + C) = (A \otimes B) + (A \otimes C)$.
- **Mixed-Product Property:** $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ (provided AC and BD are defined).

User prompt: $$A \otimes B = \begin{bmatrix} 1 \cdot B & 2 \cdot B \end{bmatrix} = \begin{bmatrix} 1 \cdot [3] & 2 \cdot [3] \end{bmatrix} = \begin{bmatrix} 3 & 6 \end{bmatrix}$$ Why not \begin{bmatrix} [3] & [6] \end{bmatrix}?

Response: In linear algebra, a 1×1 matrix with a single scalar inside it, like $[3]$, is mathematically **isomorphic** to the scalar 3 itself.

When you expand the block matrix, each sub-entry resolves to a scalar number, not a nested bracket.

Why standard matrices don't have "nested" brackets

A matrix in linear algebra is defined strictly as a 2D array of numbers:

- **A single block:** $1 \cdot B = 1 \cdot \begin{bmatrix} 3 \end{bmatrix} = \begin{bmatrix} 3 \end{bmatrix}$. This is a block containing one row and one column.
- **Combining blocks:** When you form the full block matrix $\begin{bmatrix} A_{11} & A_{12} \end{bmatrix}$, you drop the inner boundary walls of the blocks and concatenate their elements into one outer grid.

If you kept the internal brackets as $\begin{bmatrix} [3] & [6] \end{bmatrix}$, it would represent a **nested array** or a matrix whose entries are themselves matrices (a block matrix notation), rather than a standard real matrix in $\mathbb{R}^{1\times 2}$.

Dimensions prove why the brackets collapse

Recall the dimension formula for the Kronecker product:

$$\text{If } A \in \mathbb{R}^{m\times n} \text{ and } B \in \mathbb{R}^{p\times q}, \text{ then } (A \otimes B) \in \mathbb{R}^{(m\cdot p)\times(n\cdot q)}$$

For our matrices:

- A is 1×2 ($m = 1, n = 2$)
- B is 1×1 ($p = 1, q = 1$)

The resulting dimension must be:

$$\text{Rows: } 1 \cdot 1 = 1$$

$$\text{Columns: } 2 \cdot 1 = 2$$

The product $A \otimes B$ **must** be a 1×2 matrix with 1 row and 2 numeric columns: $\begin{bmatrix} 3 & 6 \end{bmatrix}$.

Writing $\begin{bmatrix} [3] & [6] \end{bmatrix}$ is just an unsimplified intermediate representation before dropping the inner block boundaries.

User prompt: Tell me about point-wise multiplication of matrices

Response: Point-wise multiplication of matrices (also known as the **Hadamard product** or **element-wise product**) multiplies corresponding entries of two matrices of the **exact same dimensions**.

Unlike standard matrix multiplication, it ignores rows-by-columns dot products and simply multiplies matching elements directly.

Mathematical Notation & Definition

The Hadamard product is represented by the open circle symbol: $\odot$ (or sometimes $\circ$).

If A and B are both $m \times n$ matrices in $\mathbb{R}^{m \times n}$, then $A \odot B$ produces an $m \times n$ matrix where each entry $(A \odot B)_{ij}$ is defined as:

$$(A \odot B)_{ij} = A_{ij} \cdot B_{ij}$$

Step-by-Step Example

Let A and B both be 2×2 matrices:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 0 \\ -1 & 2 \end{bmatrix}$$

To compute $A \odot B$, multiply each position individually:

$$A \odot B = \begin{bmatrix} 1 \cdot 5 & 3 \cdot 0 \\ 2 \cdot (-1) & 4 \cdot 2 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -2 & 8 \end{bmatrix}$$

Key Properties

- **Commutative:** $A \odot B = B \odot A$ (Unlike standard matrix multiplication, order does not matter).
- **Associative:** $(A \odot B) \odot C = A \odot (B \odot C)$.
- **Distributive over Addition:** $A \odot (B + C) = (A \odot B) + (A \odot C)$.
- **Identity Element:** A matrix full of ones (J, the all-ones matrix) acts as the identity: $A \odot J = A$.

Real-World Applications

- **Neural Networks & Deep Learning:** Used heavily in activation functions, attention mechanisms, and recurrent units like LSTMs/GRUs (e.g., gating mechanisms where a gate matrix $G \in [0, 1]$ selectively scales hidden state values).
- **Image Processing:** Applying image masks or filters. A binary mask matrix multiplied element-wise with an image tensor isolates specific regions of interest.
- **Computer Code:** In programming frameworks like NumPy or PyTorch, the standard multiplication operator `*` performs Hadamard (point-wise) multiplication, whereas `@` or `np.matmul` performs standard matrix multiplication.